Does Strengthening Rent Control Reduce Housing Quality? Evidence from New York City

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First draft: February 2026
This version: May 9, 2026

Abstract

This paper studies whether strengthening rent control reduces housing quality. I study New York's 2019 Housing Stability and Tenant Protection Act (HSTPA), which eliminated vacancy decontrol and sharply reduced landlords' ability to recover capital costs through rent increases. Using building-level panel data on housing code violations, tenant complaints, and building permits in a difference-in-differences design, I estimate the effect of HSTPA on maintenance and investment behavior. I find that housing quality declined after HSTPA: immediately hazardous violations increased by 37%, or approximately 23,000 additional violations per year, tenant complaints increased by 12%, and violations took 34% longer to resolve. Building permit activity also declined by 27%. These effects are sharply unequal: hazardous violations increased three to four times more in lower-income and predominantly non-white neighborhoods. These findings provide causal evidence that landlords reduced both routine maintenance and capital investment in response, with the costs falling disproportionately on the communities rent control is intended to protect.

JEL Codes: R31, R38, L85

Keywords: Rent control, housing quality, tenant protection, building maintenance

1 Introduction

Rising housing costs have made affordability a central policy issue in high-demand housing markets (Galster and Lee, 2021; Baum-Snow and Duranton, 2025). Policymakers have responded by expanding or strengthening rent regulation (Kholodilin, 2024).[1] One important recent reform is New York's 2019 Housing Stability and Tenant Protection Act (HSTPA), which tightened rent stabilization for roughly one million apartments. Before HSTPA, landlords could raise rents after vacancies, recover capital improvements through permanent rent increases, and eventually remove high-rent units from stabilization. HSTPA eliminated vacancy decontrol, sharply curtailed vacancy increases, and made capital-improvement recovery smaller and temporary. These provisions strengthened tenant protections, but they also weakened the link between building quality and future rental income. This paper asks whether that change caused landlords to reduce maintenance and investment in the regulated housing stock.

A central economic concern with rent regulation is that it can weaken incentives to maintain and improve housing. If landlords cannot recover maintenance and investment costs through future rents, they may reduce upkeep, allowing quality to deteriorate (Friedman and Stigler, 1946; Kholodilin, 2024). Yet this margin is difficult to study. Housing quality is multidimensional, partly hidden from researchers, and often measured through outcomes that may also reflect tenant reporting or enforcement activity. As a result, credible causal evidence on how changes in rent regulation affect the quality of continuing rental buildings remains more limited than evidence on rents, mobility, or supply.

I address this measurement challenge by assembling a building-level panel of New York City administrative records from 2017 through 2025. The data combine rent-stabilization status with HPD housing code violations, tenant complaints, violation closure times, DOB alteration permits, building characteristics, neighborhood characteristics, and mortgage records. These outcomes capture complementary dimensions of housing quality: inspector-verified hazards, tenant-reported problems, repair responsiveness, and capital investment. The richness of the data is central to the analysis because it allows me to study not just whether violations rise, but whether the broader pattern is consistent with reduced maintenance and investment.

Empirically, I compare rent-stabilized buildings to non-stabilized buildings before and after HSTPA using a difference-in-differences design. The specification includes building

[1] Recent examples include Oregon's SB 608, enacted in 2019, which limited annual rent increases to 7 percent plus CPI; California's 2019 Tenant Protection Act, which capped annual increases at 5 percent plus local CPI or 10 percent, whichever is lower; St. Paul's 2021 rent stabilization ordinance; and Montgomery County, Maryland's 2023 rent stabilization law.

fixed effects and tract-by-period, size-by-period, vintage-by-period, building-type-by-period, and baseline-violation-by-period fixed effects. Identification therefore comes from comparing stabilized and non-stabilized buildings observed in the same period and local housing market, while allowing buildings of different sizes, ages, types, and baseline violation propensities to follow separate time paths.

I find that HSTPA reduced housing quality. Immediately hazardous Class C violations increased by 1.2 violations per 100 units per 6-month period, a 37% increase relative to the stabilized-building baseline. This implies approximately 23,000 additional immediately hazardous violations per year across the city's stabilized stock. Tenant complaints increased by 12%, violations took 34% longer to resolve, and building permit activity declined by 27%. The effects are concentrated in the most serious violation category: Class C violations rise substantially, while Class A violations rise little. The combination of more serious verified violations, more tenant complaints, slower closure, and lower permit activity is difficult to reconcile with a pure reporting explanation.

The effects are also sharply unequal. Class C violations increased by 1.80 per 100 units in low-income neighborhoods, compared with 0.58 in high-income neighborhoods. In neighborhoods in the top tercile of non-white population share, violations increased by 2.33 per 100 units, compared with 0.61 in neighborhoods in the bottom tercile. These differences suggest that the quality costs of stronger rent regulation are concentrated in the communities least able to absorb them.

Financial constraints amplify the response but do not fully explain it. Buildings with pre-HSTPA mortgages show substantially larger quality declines, consistent with Seltzer (2024), who shows that financing constraints reduce maintenance in rent-stabilized apartments. But even buildings without a recorded pre-HSTPA mortgage experience significant deterioration. The effect among unlevered buildings suggests that HSTPA reduced maintenance not only by tightening cash flow for leveraged owners, but also by lowering the return to preserving quality.

This paper contributes to the literature on rent control and landlord behavior. The modern empirical literature shows that rent control affects rental supply, tenant mobility, and housing-market allocation. Diamond et al. (2019) find that San Francisco's 1994 rent-control expansion reduced rental supply through conversions and redevelopment. Autor et al. (2014) show that the end of rent control in Cambridge increased property values and investment, with spillovers to nearby units (Autor et al., 2019). Rent control also generates tenant-unit misallocation and price spillovers to the unregulated segment (Glaeser and Luttmer, 2003; Mense et al., 2023). These studies show that landlords and tenants respond to rent regulation, but they do not directly estimate whether stronger rent control causes active deterioration among buildings that remain in the rental stock.

Direct evidence on housing quality is thinner. Moon and Stotsky (1993) use a structural model to show that greater rent suppression in New York City was associated with faster deterioration. Sims (2007) finds that decontrol in Massachusetts reduced maintenance problems, though primarily for minor deficiencies. Breidenbach et al. (2022) find that Germany's rent brake reduced advertised listing quality, but the rent effects faded within about a year. Baye and Dinger (2024) show that Germany's rent brake compressed returns on regulated properties and shifted investment toward exempt new construction. The closest paper is Seltzer (2024), who uses building-level violation data in New York City and shows that a 2011 reduction in cost recovery for unit improvements increased violations among highly leveraged buildings.

The closest HSTPA-specific evidence is descriptive. NYU Furman Center (2021) examine short-run trends in complaints, violations, and alteration filings after HSTPA and find little immediate change in complaints or violations, though alteration filings declined after an initial spike. More recently, NYU Furman Center (2026) descriptively document deteriorating operating margins and larger post-2021 increases in code violations among legacy buildings with at least 90 percent rent-stabilized units. These reports show that the financial and physical condition of highly stabilized buildings is a first-order policy concern, but they do not isolate HSTPA's causal effect from enforcement, reporting, or post-pandemic cost shocks.

My contribution is to provide causal evidence on HSTPA's longer-run quality effects using building-level administrative data. I distinguish routine maintenance outcomes from capital investment, show that effects persist over six years rather than fading quickly, and document that quality deterioration is concentrated in lower-income and predominantly non-white neighborhoods. The results imply that strengthening rent regulation can reduce both maintenance and investment in the existing rental stock, and that these costs may fall disproportionately on tenants the policy is intended to protect.

The rest of the paper proceeds as follows. Section 2 describes HSTPA and the investment incentives it changed. Section 3 describes the building-level panel. Section 4 presents the empirical strategy. Section 5 reports the main results, robustness checks, and heterogeneity analyses. Section 6 concludes.

2 Policy Background

2.1 Rent Stabilization in New York City

New York City's rent stabilization system covers approximately one million apartments, primarily in buildings with six or more units built before 1974.[2] Unlike strict rent control, which freezes rents at a fixed level, rent stabilization allows annual rent increases set by the Rent Guidelines Board, typically 1–3% per year for lease renewals. Arnott (1995) calls this "second-generation" rent control: landlords can recover operating cost increases, but rent growth is capped.

Prior to HSTPA, landlords could also increase rents through several additional mechanisms:

- **Vacancy bonuses:** Upon tenant turnover, landlords could increase rents by approximately 20%, reflecting the higher costs associated with unit turnover and the opportunity to reset rents closer to market levels.
- **Major Capital Improvements (MCI):** Landlords who made building-wide improvements (boilers, roofs, elevators, windows) could recover the cost through permanent rent increases of up to 6% of the legal rent annually, spread across all units in the building.
- **Individual Apartment Improvements (IAI):** Landlords who renovated individual units could increase the legal rent by 1/40th of the improvement cost (effectively a 2.5-year payback period) for units with rent above $2,500, or 1/60th for units below that threshold.
- **High-rent vacancy decontrol:** Units could exit stabilization entirely when the legal rent exceeded approximately $2,774 per month upon vacancy, or when rent exceeded $2,774 and household income exceeded $200,000 for two consecutive years.

These provisions gave landlords multiple ways to raise rents and, through vacancy decontrol, to eventually exit the system entirely. The result was that landlords had strong incentives to invest: capital improvements were recoverable through MCI and IAI increases, tenant turnover brought vacancy bonuses and moved units toward decontrol, and maintaining buildings in good condition attracted higher-income tenants.

[2]Exact counts vary depending on data source and year. The NYC Rent Guidelines Board estimated 966,000 stabilized units in 2018; see `https://rentguidelinesboard.cityofnewyork.us/resources/rent-stabilized-building-lists/`.

2.2 The Housing Stability and Tenant Protection Act (2019)

On June 11, 2019, the New York State Legislature passed the Housing Stability and Tenant Protection Act; Governor Cuomo signed it three days later. The law made the following major changes to rent stabilization:

1. **Eliminated high-rent vacancy decontrol:** Units can no longer exit stabilization regardless of rent level or tenant income. This change effectively closes the main exit option that landlords had previously used to transition buildings to market-rate status.

2. **Capped vacancy bonuses:** Reduced from approximately 20% to a maximum of 0–5%, depending on the time since the last vacancy increase. For units that had not received a vacancy increase in eight or more years, a 5% bonus is permitted; otherwise, no vacancy bonus is allowed.

3. **Reformed MCI increases:** Reduced annual rent increases from 6% to 2% of the legal rent, and made increases temporary (amortized over 30 years) rather than permanent. This change reduces the present value of MCI cost recovery by approximately 50–70%, depending on discount rates and assumptions about future rent growth. Additionally, MCIs are now prohibited for buildings where 35% or fewer of units are rent-stabilized.

4. **Capped IAI increases:** Limited to $15,000 over any 15-year period, with increases made temporary rather than permanent. Previously, there was no cap on total IAI spending, and increases were permanent additions to the legal rent.

5. **Eliminated preferential rent resets:** Landlords can no longer raise preferential rents (voluntarily reduced rents below the legal maximum) to the legal maximum upon vacancy. This change reduces the incentive for landlords to offer preferential rents and may affect tenant retention decisions.

6. **Restricted condo conversions:** Increased the tenant consent threshold for converting rental buildings to condominiums from 15% to 51%, making conversions substantially more difficult.

7. **Limited owner reclaim:** Restricted the number of units an owner can reclaim for personal use to one per building, reducing a pathway landlords had used to remove stabilized units from the rental market.

Taken together, these provisions tightened the rent control system, limited landlords' ability to exit rent stabilization, raise rents in response to capital investment, or convert rental units toward owner-occupation. This paper asks whether these changes led landlords to reduce maintenance and investment in their rental properties.

2.3 Investment Incentives and Empirical Predictions

The provisions described above reduced the expected return to maintaining and improving rent-stabilized buildings. Before HSTPA, landlords could recover part of the cost of quality improvements through vacancy bonuses, MCI and IAI rent increases, and eventual vacancy decontrol. HSTPA sharply weakened or eliminated these channels, reducing the link between building quality and future rental income.

These changes generate two empirical predictions. First, HSTPA should reduce capital investment in stabilized buildings, since landlords can recover less of the cost of renovations through rent increases or decontrol. I measure this margin using building permits, which capture renovation and alteration work requiring Department of Buildings approval.

Second, HSTPA should worsen routine maintenance. Lower expected returns to preserving building quality weaken incentives to prevent deterioration or repair problems quickly. I measure this margin using serious housing code violations, tenant complaints, and the time required to close violations.

3 Data

3.1 Data Sources and Variables

I assemble administrative data from multiple New York City agencies to measure housing quality, investment, and rent stabilization status at the building level. I link datasets using the Borough-Block-Lot (BBL) identifier, a 10-digit code that uniquely identifies every tax lot in New York City.

Building characteristics. The Primary Land Use Tax Lot Output (PLUTO) database, published by the Department of City Planning and available through NYC Open Data, provides the universe of tax lots and their characteristics. [3] I use PLUTO to obtain total residential units, year built, building class, census tract, and lot area. PLUTO defines the sample frame: I begin with all lots and apply the restrictions described below.

Housing quality outcomes. Data on violations and complaints come from the Department of Housing Preservation and Development (HPD), accessed through NYC Open Data. HPD enforces the New York City Housing Maintenance Code. Tenants file complaints

[3] NYC Open Data is the city's public data portal, available at `https://opendata.cityofnewyork.us/`.

through 311 (phone or online) describing conditions such as lack of heat, water leaks, or pest infestations. HPD inspectors then visit the building to verify reported conditions. If a violation of the Housing Maintenance Code is confirmed, it is classified by severity: Class A (non-hazardous), Class B (hazardous), or Class C (immediately hazardous). Class B violations, which include defective plumbing, inadequate lighting in public areas, broken window guards, and missing smoke detectors, must be corrected within 30 days. Class C violations, which include lead paint hazards, no heat or hot water, pest infestations, and structural defects, must be corrected within 24 hours.

I focus on Class C violations as the primary outcome because they represent the most serious maintenance failures. I also examine Class B (hazardous) violations. Violations are closed when either an HPD inspector verifies the condition has been corrected or the landlord submits a certificate of correction. I measure days to close as a proxy for landlord responsiveness.

HPD complaint data provide a complementary measure. Unlike violations, which require inspector verification, complaints represent tenant-reported problems directly, capturing tenant experience without necessarily depending on enforcement intensity.

Investment outcomes. Building permit data come from the Department of Buildings (DOB), also accessed through NYC Open Data. NYC law requires permits for significant construction work: alterations affecting building use or occupancy, structural modifications, and major plumbing or electrical work. I focus on alteration permits (types A1, A2, A3), which capture renovations requiring regulatory approval. I combine data from both DOB's traditional filing system and DOB NOW, the online portal launched in 2016 that now handles the majority of filings.[4]

Routine maintenance (painting, plaster repair, replacing fixtures, minor repairs) does not require permits. This distinction is central to interpretation: permit activity captures major capital investment, while violations and complaints capture routine maintenance. I normalize permit counts to per-100-units, consistent with the violation and complaint outcomes.

For descriptive analysis of the MCI cost-recovery mechanism, I use Major Capital Improvement application data from the NYCDB project.[5] MCIs are building-wide improvements (boilers, roofs, elevators) for which stabilized building landlords can seek rent increases.

Treatment variables. I collect data on rent-stabilization status from two sources. First, I collect the list of buildings with any rent-stabilized units from the Division of Housing and

[4]Using only the traditional system would show a spurious decline after 2016 as filings shifted to the online portal.

[5]The MCI application records were originally obtained from HCR/DHCR through a FOIL request by Winnie Shen and distributed through NYCDB; see `https://github.com/nycdb/nycdb`.

Community Renewal (DHCR) registry, published by the NYC Rent Guidelines Board as PDF lists of stabilized buildings (BBLs) by borough.[6] I extract BBLs from these PDFs and match to the PLUTO database. Using the 2017 registry ensures that treatment classification is determined before HSTPA's passage in June 2019, avoiding any concern that post-treatment changes in stabilization status could bias the estimates.

For robustness, I also collect data on the share of units that were rent-stabilized in each building as of 2018, which I use as a continuous treatment measure. These data are downloaded from the NYCDB project, and were originally compiled from NYC Department of Finance (DOF) tax bills.[7]

Building-level leverage. To test whether effects operate through a debt channel, I construct building-level leverage measures from the Automated City Register Information System (ACRIS), NYC's public database of property transactions and mortgage recordings.[8] For each building, I identify the most recent mortgage origination before HSTPA and compute the ratio of mortgage amount to 2018 assessed value from PLUTO. This measure captures relative leverage across buildings but does not represent true current loan-to-value ratios, since I observe origination amounts rather than outstanding balances. The median mortgage-to-assessed ratio in my sample is 1.76.[9] I split buildings with pre-HSTPA mortgages into terciles by this ratio; buildings without pre-HSTPA mortgage records form the reference group.

Landlord identification. To test for portfolio spillovers across a landlord's stabilized and non-stabilized buildings, I link buildings to individual owners using HPD Multiple Dwelling Registration data.[10] NYC law requires owners of buildings with three or more residential units to register annually with HPD, providing contact information including the name of the owner or head officer. I extract standardized owner names and link buildings that share common ownership. Of the 53,643 buildings in my sample, 52,579 (98%) match to an owner; of these, 1,880 unique owners (7.5%) have "mixed portfolios" containing both stabilized and non-stabilized buildings. Appendix C provides details on the name standard-

[6] Available at `https://rentguidelinesboard.cityofnewyork.us/resources/rent-stabilized-building-lists/`. I use the 2017 registry, archived at `https://github.com/austensen/dhcr-rgb-buildings`.

[7] Data originally compiled by John Krauss from DOF tax bills; see `https://github.com/nycdb/nycdb`.

[8] ACRIS data available through NYC Open Data. I use mortgage documents (type MTGE) recorded before June 2019.

[9] NYC assesses multifamily rental buildings (Class 2) at 45% of market value, so a mortgage-to-assessed ratio of 1.76 corresponds to an approximate market loan-to-value ratio of 79% at origination. See NYC Department of Finance, *Definitions of Property Assessment Terms*, `https://www.nyc.gov/site/finance/property/definitions-of-property-assessment-terms.page`.

[10] HPD Registration Contacts (`feu5-w2e2`) and Multiple Dwelling Registrations (`tesw-yqqr`) from NYC Open Data.

ization and portfolio classification.

Neighborhood characteristics. For heterogeneity analysis, I use tract-level median household income and non-white population share from the 2015–2019 American Community Survey 5-year estimates, accessed via the Census API.

3.2 Sample and Summary Statistics

The analysis sample includes buildings with 6 or more residential units, constructed before 1990, observed from June 2016 through December 2025. I restrict to older, larger buildings to focus on the multifamily rental stock where rent stabilization is prevalent and to ensure common support between stabilized and non-stabilized buildings. I exclude NYCHA public housing and buildings receiving LIHTC, HUD-assisted, or Mitchell-Lama subsidies, which face federal or state affordability requirements distinct from rent stabilization.[11]

Table 1 summarizes the sample construction, showing how the final analysis sample is derived from the universe of NYC properties. The sample begins with all lots in the PLUTO database, then applies restrictions for multifamily buildings (6+ units), valid year built, construction before 1990, and available community district information. NYCHA public housing and subsidized housing (LIHTC, HUD-assisted, Mitchell-Lama) are excluded because these buildings face distinct regulatory regimes.

Table 2 presents summary statistics separately for rent-stabilized and non-stabilized buildings. Stabilized buildings average 30 units compared to 32 for non-stabilized buildings, and both groups were built around 1920–1922 on average. However, stabilized buildings have substantially higher baseline violation rates (0.17 Class C violations per month vs. 0.06) and complaint rates (0.56 vs. 0.18 per month). Figure 1 shows the geographic distribution: rent-stabilized buildings are concentrated in Manhattan, the Bronx, and parts of Brooklyn and Queens, with substantial overlap between stabilized and non-stabilized buildings within neighborhoods.

Both stabilized and non-stabilized buildings in the sample are predominantly prewar construction (92% and 88%, respectively), with similar distributions across walk-up and elevator building types. The non-stabilized buildings are largely prewar buildings that either decontrolled prior to HSTPA through vacancy decontrol or were never enrolled in stabilization. Only 4.6% of non-stabilized buildings were constructed between 1974 and 1990 (after stabilization began). I exclude condominiums from the sample because individual unit owners, rather than a single landlord, make maintenance decisions; results including condos are nearly identical.

[11] I identify subsidized buildings using the HPD Affordable Building dataset from NYC Open Data.

4 Empirical Strategy

The goal is to estimate the causal effect of HSTPA on building outcomes by comparing rent-stabilized buildings (treated) to non-stabilized buildings (control) before and after June 2019. The challenge is that stabilized and non-stabilized buildings differ systematically. As Table 2 shows, stabilized buildings have nearly three times as many Class C violations per month at baseline (0.17 vs. 0.06) and over three times as many complaints (0.56 vs. 0.18). Simple comparisons would conflate HSTPA effects with pre-existing differences in building quality.

I address this with building fixed effects and five sets of group-by-period fixed effects. Building fixed effects absorb all time-invariant characteristics, so identification comes from within-building changes over time. Tract × period fixed effects absorb any neighborhood-level shock, including gentrification, shifts in local housing demand, and changes in enforcement intensity, so that identification comes from within-neighborhood comparisons of stabilized and non-stabilized buildings experiencing the same local conditions. Size × period, decade × period, and building class × period fixed effects allow buildings with different physical characteristics to follow separate trends, ensuring comparisons are among physically similar buildings.[12]

Finally, baseline outcome tercile × period fixed effects address the fact that stabilized and non-stabilized buildings differ substantially in baseline outcome levels (Table 2), and buildings at different quality levels likely evolve on different trajectories regardless of treatment. I compute baseline terciles from the three-year pre-HSTPA average separately for each outcome and interact them with period fixed effects, allowing each quality tier to follow its own path. Because terciles are computed from a multi-year average rather than a single lagged value, they primarily capture persistent building characteristics rather than transitory fluctuations. Thus, identification comes from comparing stabilized and non-stabilized buildings observed in the same period and local tract, while allowing buildings of different sizes, vintages, types, and baseline violation propensities to follow separate time paths. Table A2 in Appendix A.2 demonstrates the importance of this specification: without controlling for baseline outcomes, pre-trends are severely violated, but adding baseline-tercile × time fixed effects achieves clean pre-trends with no pre-period coefficient exceeding conventional significance thresholds. I estimate:

$$Y_{it} = \alpha_i + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[t = k] \cdot \text{Stabilized}_i + \gamma^{\text{tract}}_{c(i),t} + \gamma^{\text{size}}_{s(i),t} + \gamma^{\text{decade}}_{d(i),t} + \gamma^{\text{type}}_{b(i),t} + \gamma^{\text{baseline}}_{q(i),t} + \varepsilon_{it} \quad (1)$$

[12] Size bins are 6–10, 11–20, 21–50, 51–100, and 100+ units. Decade bins span pre-1920 through the 1990s. Building classes are walk-up apartments, elevator apartments, and mixed-use buildings.

where Y_{it} is the outcome for building i in period t, α_i are building fixed effects, and the γ terms are the five sets of group-by-period fixed effects (indexed by tract c, size bin s, decade d, building type b, and baseline tercile q).[13] The coefficients β_k capture the differential change in outcomes for stabilized buildings in period k relative to the reference period ($k = -1$, January–June 2019). Standard errors are clustered at the building level to account for serial correlation within buildings.

I also report pooled difference-in-differences estimates:

$$Y_{it} = \alpha_i + \beta \cdot \text{Post}_t \cdot \text{Stabilized}_i + \gamma^{\text{tract}}_{c(i),t} + \gamma^{\text{size}}_{s(i),t} + \gamma^{\text{decade}}_{d(i),t} + \gamma^{\text{type}}_{b(i),t} + \gamma^{\text{baseline}}_{q(i),t} + \varepsilon_{it} \tag{2}$$

where $\text{Post}_t = \mathbf{1}[t \geq 0]$ indicates periods after HSTPA. The coefficient β captures the average post-period treatment effect.

For count outcomes (violations, complaints), I normalize to per-100-units and weight by building size. This provides a unit-weighted interpretation: effects reflect the experience of the average housing unit rather than the average building. Appendix A.1 reports unweighted count-based results, which are similar.

I aggregate time into 6-month bins rather than months for three reasons. First, violations and complaints are relatively rare at the building level, so monthly data would be noisy. Second, 6-month bins smooth over seasonal patterns (e.g., heating complaints concentrate in winter). Third, 6-month bins align naturally with HSTPA's timing: the law passed on June 14, 2019, so I assign June 2019 to the final pre-period bin (January–June 2019) and define July–December 2019 as the first post-period bin.

As a robustness check, I replace the binary treatment indicator with the share of units that are rent-stabilized, allowing for dose-response effects where fully stabilized buildings may respond more than partially stabilized buildings.

The key identifying assumption is parallel trends: conditional on the fixed effects, outcomes in stabilized and non-stabilized buildings would have evolved similarly absent HSTPA. The baseline-tercile × period fixed effects implement a conditional parallel trends assumption—they allow different baseline groups to follow different trajectories, but do not mechanically force flat pre-trends. Within each baseline group, differential trends between stabilized and non-stabilized buildings would still appear in the event study coefficients. The event study provides an indirect test: if pre-period coefficients are close to zero, the groups were on parallel trajectories before HSTPA. As I show in Section 5, the full set of fixed effects is essential for achieving parallel pre-trends.

A further assumption is no anticipation: landlords did not adjust behavior before HSTPA's

[13] Size bins are 6–10, 11–20, 21–50, 51–100, and 100+ units. Decade bins span pre-1920 through the 1990s. Building types follow NYC Department of Finance classifications (walk-up, elevator, etc.).

passage. Although the law was publicly debated in the months preceding June 2019, the pre-period event study coefficients are close to zero across all outcomes, suggesting limited anticipatory response on the maintenance margin. To the extent that landlords anticipated HSTPA by accelerating capital investment before the law took effect—as suggested by the spike in MCI applications in June 2019—this would mean stabilized buildings entered the post-period in better condition than they otherwise would have, attenuating the estimated effects.

Furthermore, treatment status could not have been altered in anticipation or response to the policy: classification is determined from the 2017 DHCR registry, fixed before HSTPA's passage.

A second concern is SUTVA: if landlords who own both stabilized and non-stabilized buildings reallocate resources between them after HSTPA, the control group is contaminated. Section 5 addresses this by comparing non-stabilized buildings owned by mixed-portfolio landlords to those owned by all-market landlords; reallocation would cause the former to improve relative to the latter.

5 Results

5.1 Main Results

Violations. Housing code violations provide a direct measure of building conditions, with Class C violations representing immediately hazardous conditions. Figure 2 presents the event study. The pre-period coefficients are close to zero and statistically insignificant. Violations increase after HSTPA, with coefficients becoming positive and significant starting in the first post-period (July–December 2019). The effect persists and grows over time.

Table 3 reports the pooled difference-in-differences estimates. Class C violations increased by 1.2 violations per 100 residential units per 6-month period, a 37% increase relative to the pre-HSTPA baseline of 3.2 Class C violations per 100 units for stabilized buildings. Class B violations increased by 0.70 per 100 units (8% of baseline). The time to close violations increased by 35 days (34% of baseline), indicating longer delays before cited problems are resolved. This reduced-form effect may reflect slower landlord remediation, shifts in violation mix or severity, and/or changes in enforcement processes.

Complaints. Figure 3 presents the event study for tenant complaints. Complaints represent tenant-reported problems rather than inspector-identified violations, providing a complementary measure of building quality that does not depend on enforcement intensity.

Table 3 shows that HPD complaints increased by 1.3 per 100 units per 6-month period

(12% of baseline). These findings address the concern that violation results might reflect changes in enforcement intensity: complaints are filed by tenants regardless of inspection activity.

HSTPA may have increased tenant willingness to report problems, which could explain some of the violation increase even absent actual quality decline. However, several considerations suggest this is not the primary driver. First, Class C violations are immediately hazardous conditions (no heat, gas leaks, severe vermin) that landlords must address within 24 hours; tenants are unlikely to leave such conditions unreported in any case. Second, violation increases are largest for the most serious violation class and smallest for the least serious: Class C violations increased 37%, Class B (hazardous, 30-day deadline) increased 8%, and Class A (non-hazardous) increased just 3% (Appendix B.2). If the results reflected increased reporting of marginal conditions, we would expect the opposite pattern. Third, the permit decline provides corroborating evidence that does not depend on tenant behavior: permits reflect landlord decisions filed with the Department of Buildings, and the 27% decline cannot be explained by changes in reporting. Fourth, complaints—which most directly capture tenant reporting behavior—increased 12%, substantially less than the 37% increase in inspector-verified Class C violations. If increased reporting were the primary channel, we would expect complaints to increase more than violations, not less. Together, these patterns suggest real reductions in investment rather than solely increased reporting.

Permits. Figure 4 presents the event study for building permits. Alteration permits are required for significant construction work including structural modifications, major plumbing, and electrical work, and thus capture capital investment rather than routine maintenance. Permit activity declined by -0.24 permits per 100 units per 6-month period following HSTPA, a 27% decline relative to the baseline of 0.87 permits per 100 units.

The permit decline is smaller in percentage terms than the violation and complaint increases, but still economically meaningful.

Separately, MCI applications—which allow landlords to recover the cost of building-wide improvements through rent increases—collapsed 81% immediately upon HSTPA's passage. Because MCIs are available only for rent-stabilized buildings, there is no valid control group, so this is descriptive evidence only. Appendix B.1 presents the time series.

5.2 Robustness

Fixed effects specifications. Table A2 in Appendix A.2 estimates specifications with progressively richer fixed effects structures for Class C violations. All specifications produce positive point estimates, but simpler specifications exhibit substantial pre-trend concerns.

With only building fixed effects, pre-period coefficients are highly significant. Adding neighborhood and building characteristic controls progressively reduces pre-trend violations, but only the full specification with baseline-tercile $\times$ time fixed effects achieves clean pre-trends.

One concern with matching on baseline outcomes relates to mean reversion: one might worry that non-stabilized buildings in high-violation terciles are there due to transitory shocks and would naturally revert toward lower violation rates, generating a spurious positive effect. Appendix A.5 addresses this by estimating event studies separately within each baseline tercile. Pre-trends are flat within every tercile, and effects emerge only after HSTPA. This rules out differential mean reversion as a confound.

Treatment intensity. If HSTPA caused the observed decline in maintenance quality, effects should scale with treatment intensity. Figure 5 tests this by comparing the binary treatment (any stabilization) with a continuous measure (share of units stabilized).

Both specifications show flat pre-trends followed by divergence after HSTPA. The continuous specification yields a larger point estimate, consistent with dose-response: fully stabilized buildings experience larger effects than partially stabilized buildings. Table A3 in Appendix A.3 confirms that this pattern holds across all outcomes. As a further check, Table A4 restricts the sample to stabilized buildings only, comparing buildings with more or fewer stabilized units. Effects remain significant across outcomes here as well.

Portfolio spillovers (SUTVA). A potential threat to identification arises if landlords who own both stabilized and non-stabilized buildings reallocate resources toward their non-stabilized properties after HSTPA, contaminating the control group.

To test for this, I link buildings to owners using HPD registration data (see Appendix C for details). I identify 1,880 "mixed-portfolio" landlords who own both stabilized and non-stabilized buildings, covering 24,810 buildings. I then compare non-stabilized buildings owned by mixed-portfolio landlords to those owned by all-market landlords. Restricting the sample to non-stabilized buildings only, I estimate:

$$y_{it} = \beta \cdot \text{Mixed}_i \times \text{Post}_t + \alpha_i + \gamma_{g(i),t} + \varepsilon_{it} \tag{3}$$

where $\text{Mixed}_i = 1$ if building i is owned by a landlord who also owns stabilized buildings. If reallocation were occurring, we would expect $\beta < 0$ (mixed-portfolio non-stabilized buildings improving after HSTPA).

Table 4 presents the results. If landlords were reallocating resources toward their non-stabilized properties, we would expect a *negative* coefficient (mixed-portfolio buildings improving relative to all-market buildings). Instead, the coefficient is positive and statistically insignificant, providing no evidence of resource reallocation. As an additional check, I reestimate the main specification excluding all mixed-portfolio buildings entirely; the effect remains positive and significant. These results suggest that SUTVA violations through portfolio reallocation are not driving the main findings.

COVID-19. The post-period includes the COVID-19 pandemic, which disrupted HPD inspections, tenant complaint behavior, and construction activity. Several features of the research design address potential bias from COVID-related disruptions. First, the tract $\times$ period fixed effects absorb any citywide or neighborhood-level disruptions that affected all buildings equally. The identifying assumption requires only that COVID did not differentially affect stabilized versus non-stabilized buildings within the same neighborhood, a plausible assumption since inspection protocols applied uniformly. Second, as a robustness check, I exclude the COVID period entirely (January 2020 through December 2021). Results are similar across all outcomes (Appendix Table A5).

Sample composition. The main sample excludes condominiums, which have different maintenance incentive structures than rental buildings (individual unit owners make decisions rather than a single landlord). Including condos in the sample yields nearly identical results, confirming that the findings are robust to this sample restriction.

5.3 Heterogeneity

The aggregate effects may mask important variation across neighborhoods. To explore this, I estimate the main specification separately by borough and by neighborhood characteristics, using tract-level median household income and non-white population share from the 2015–2019 American Community Survey.

Table 5 presents pooled difference-in-differences results for Class C violations. Effects vary substantially across boroughs: the Bronx shows the largest increase (+2.45 violations per 100 units), more than twice Manhattan's effect (+1.02). Brooklyn (+1.12) and Queens (+0.98) fall in between.

Effects also concentrate in disadvantaged neighborhoods. Splitting tracts into terciles by median household income, the effect in low-income neighborhoods (+1.80) is more than three times larger than in high-income neighborhoods (+0.58). Similarly, neighborhoods in the top tercile of non-white population share show effects nearly four times larger (+2.33) than those in the bottom tercile (+0.61). To formally test these differences, I estimate pooled specifications interacting treatment with tercile indicators. The difference between high and low non-white neighborhoods is statistically significant (t = 7.6), while middle and low non-white neighborhoods show similar effects (t = 2.4). For income, both low-income (t = 6.6) and middle-income (t = 5.2) neighborhoods show significantly larger effects than high-income neighborhoods. These tests confirm that the concentration of effects in disadvantaged neighborhoods is not merely descriptive.

The concentration of effects in lower-income and predominantly non-white neighborhoods could reflect landlord financial constraints, discriminatory maintenance behavior, differences in enforcement intensity, or other factors. Regardless of mechanism, the pattern suggests that the policy's unintended consequences fall disproportionately on disadvantaged communities.

The main results show that HSTPA reduced housing quality on average. I next ask whether these effects are larger for financially constrained buildings. If reduced expected returns are the central mechanism, landlords with less financial slack may cut maintenance more sharply because they have less ability to absorb the decline in expected returns. I test this using pre-HSTPA mortgage leverage.

Table 6 tests whether effects vary with financial capacity using the mortgage-to-assessed ratio described in Section 3.[14] The results show effects substantially larger for buildings with any mortgage: unlevered buildings show an increase of 0.59 Class C violations per 100 units, while buildings with mortgages show larger effects: low leverage (+1.10), middle leverage (+2.09), and high leverage (+1.64). The interaction terms comparing each leverage tercile to the no-mortgage group are all significant. The relationship is not strictly monotonic—middle-leverage buildings show the largest effects, possibly reflecting measurement error in the leverage proxy (origination amounts rather than outstanding balances) or the fact that the most highly leveraged buildings may already have been in financial distress before HSTPA. Effects are two to three times larger among leveraged buildings, though the effect among unlevered buildings (+0.59) remains statistically significant and economically meaningful.

This finding engages directly with Seltzer (2024), who shows that highly leveraged buildings experience larger quality declines following rent control. Consistent with his results, I find that leverage matters. But the effect exists even among unlevered buildings, which his identification strategy cannot address since it conditions on mortgaged properties. The pattern suggests that HSTPA reduced quality through the return channel (reduced expected rents lowering optimal maintenance), with leverage substantially amplifying the response.

6 Conclusion

This paper studies how strengthening rent control affects housing quality. I use New York's 2019 Housing Stability and Tenant Protection Act, which sharply reduced landlords' ability to raise rents and recover capital costs, and compare outcomes in rent-stabilized and

[14] This proxy is based on the most recent pre-HSTPA mortgage origination amount, not outstanding balances, so it captures relative leverage across buildings rather than true loan-to-value ratios.

non-stabilized buildings. I find that hazardous violations per 100 units increased by 37%, complaints increased by 12%, and violations took 34% longer to resolve. Building permit activity also declined by 27%. These effects persist over the six-year study period. Effects concentrate in lower-income and predominantly non-white neighborhoods, where violations increased three to four times more than in higher-income, predominantly white areas. These findings suggest that strengthening rent control reduced both routine maintenance and capital investment, with the largest effects in the most disadvantaged communities.

A full welfare analysis of HSTPA would need to weigh the costs documented here against the benefits to tenants of stronger rent protections and reduced displacement risk. The increase in hazardous violations represents a real cost to tenant wellbeing, but so does rent instability and the threat of eviction through decontrol. This paper does not resolve which effect dominates. What it does establish is that quality deterioration is a real and quantitatively meaningful cost of strengthening rent control, one that operates even when the supply margin is constrained by other factors. This cost should be weighed explicitly in policy design. Potential mitigations might include enhanced code enforcement, tax incentives for building maintenance, reformed capital cost recovery mechanisms, or targeted subsidies for buildings in disadvantaged neighborhoods where effects concentrate. One limitation is that I do not observe maintenance expenditures directly. Analysis of building-level leverage from mortgage records shows that effects are two to three times larger among leveraged buildings, though unlevered buildings also show significant quality declines.

These results have immediate policy relevance. New York City is currently debating a rent freeze on the city's one million rent-stabilized apartments, and rent control is under active debate in many other cities. The findings here suggest that such policies, absent countervailing measures, could accelerate the quality decline documented in this paper. More broadly, the results suggest that municipalities considering stronger rent control should anticipate potential quality effects and consider complementary measures to address them. The concentration of effects in disadvantaged neighborhoods underscores that absent such measures, the costs may fall disproportionately on the communities rent control is intended to protect.

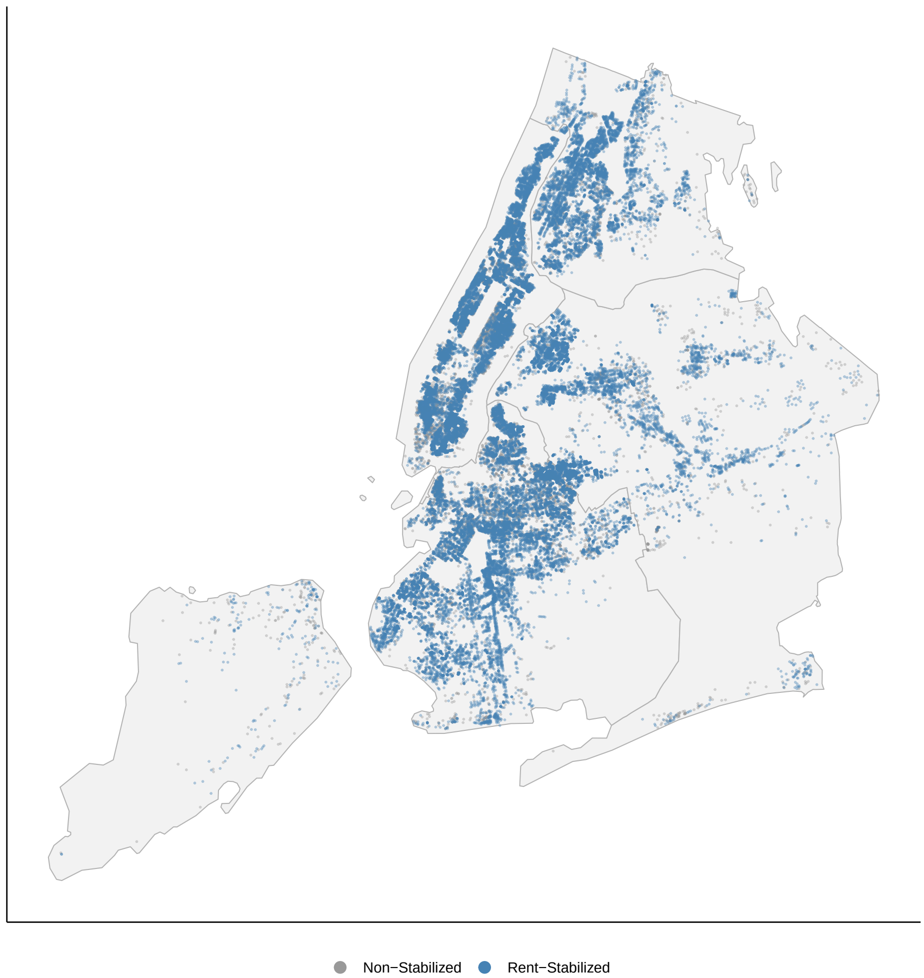
Figures and Tables

Table 1: Sample Construction

Sample Restriction	Buildings	Dropped
All NYC lots (PLUTO)	858,284	—
Multifamily ($\geq$6 units)	68,040	790,244
Valid year built	67,779	261
Built 1990 or earlier	56,968	10,811
Excl. NYCHA & subsidized	56,284	684
Excl. condos (R-class)	53,643	2,641

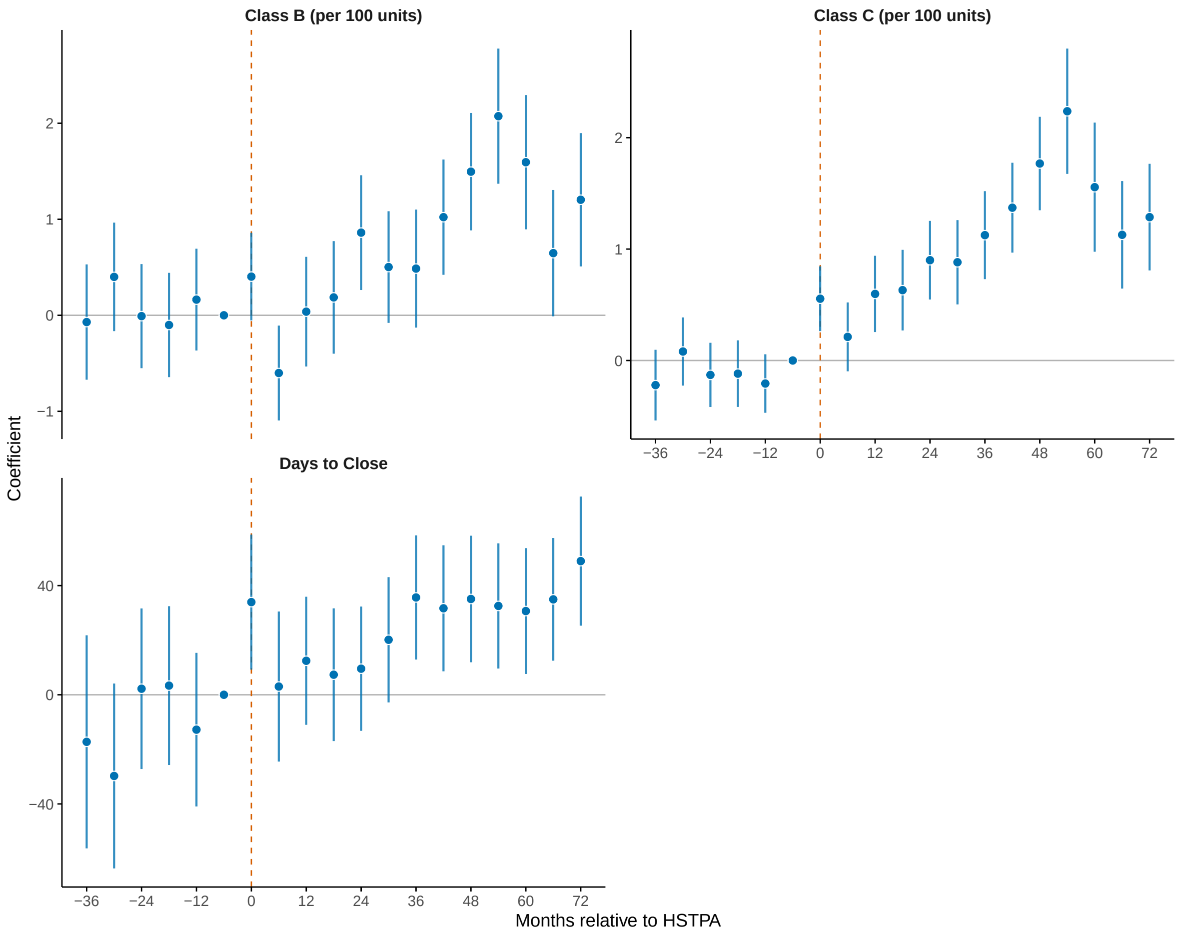
Note: Table shows the number of buildings remaining after each sample restriction. "Dropped" indicates the number of buildings excluded at each step. NYCHA refers to New York City Housing Authority public housing. Subsidized housing includes LIHTC, HUD-assisted, and Mitchell-Lama buildings.

Figure 1: Geographic Distribution of Sample Buildings by Rent Stabilization Status



Note: Blue points indicate rent-stabilized buildings (38,992); gray points indicate non-stabilized buildings (14,651), as classified by the 2017 DHCR registry. Building locations are based on PLUTO lot centroids.

Figure 2: Event Study: Effect of HSTPA on Housing Code Violations



Note: Sample consists of 53,643 NYC buildings with six or more residential units built before 1990, excluding NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Figure plots coefficients β_k on interactions between an indicator for rent-stabilized buildings and 6-month period indicators, relative to January–June 2019 (period $k = -1$). Count outcomes (Class C, Class B) are normalized to per-100-units and weighted by building size. Specification includes building fixed effects plus tract×period, size-bin×period, decade×period, building-type×period, and baseline-tercile×period fixed effects. Vertical dashed line indicates HSTPA passage (June 2019). Error bars show 95 percent confidence intervals. Standard errors are clustered at the building level.

Table 2: Summary Statistics

	Stabilized		Non-Stabilized	
	Mean	SD	Mean	SD
Panel A: Building Characteristics				
Total Units	29.9	67.0	31.6	155.1
Year Built	1922	19	1920	25
Panel B: Pre-Period Outcomes (monthly)				
Class C Violations	0.168	0.456	0.059	0.296
Class B Violations	0.475	1.213	0.156	0.755
HPD Complaints	0.555	2.069	0.177	1.014
DOB Alterations	0.041	0.139	0.059	0.168
N Buildings	38,992		14,651	

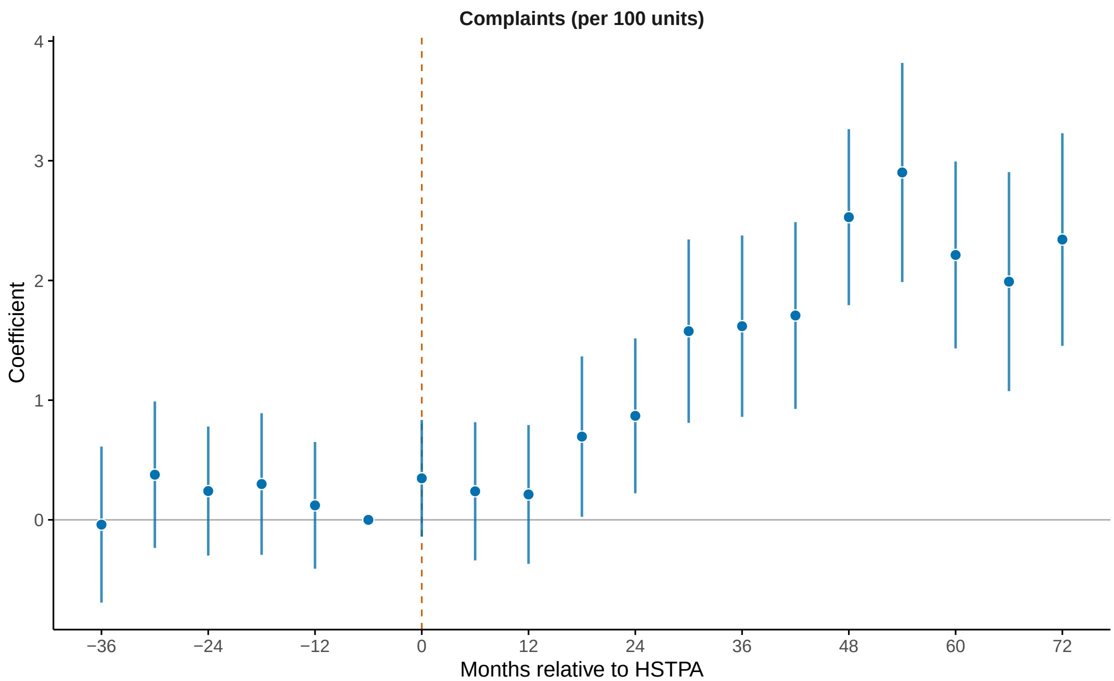
Note: Sample consists of 53,643 NYC buildings with six or more residential units built before 1990, excluding NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Rent stabilization status is determined from the 2017 DHCR registry. Violation and complaint data are from NYC HPD Open Data; permit data from NYC DOB Open Data. Outcome variables are measured at the building-month level during the pre-HSTPA period (June 2017 through May 2019). Class C violations are immediately hazardous conditions.

Table 3: Effect of HSTPA on Building Outcomes

	Class C	Class B	Complaints	Days to Close	Permits
Stabilized × Post	1.195***	0.6985***	1.314***	34.52***	-0.2361***
	(0.1114)	(0.1822)	(0.2309)	(7.122)	(0.0294)
Baseline mean	3.24	9.27	11.04	100.18	0.87
Buildings	53,642	53,642	53,642	34,126	53,642
Observations	1,019,198	1,019,198	1,019,198	207,720	1,019,198

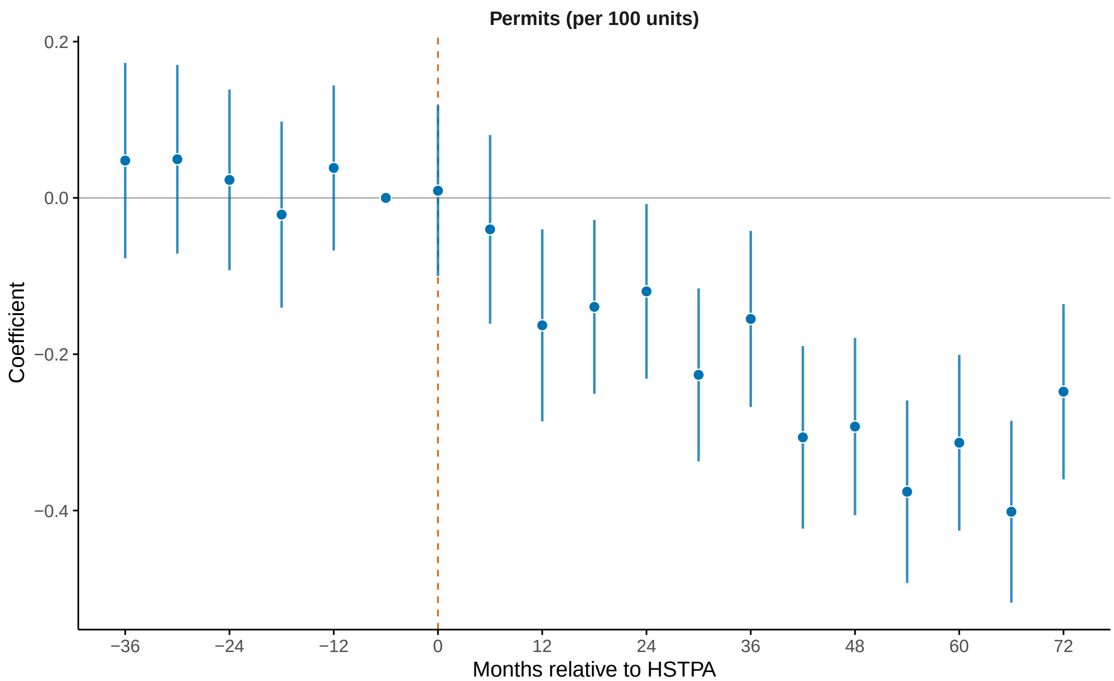
Note: This table tests whether HSTPA increased housing code violations in rent-stabilized buildings. Sample consists of 53,643 NYC buildings with six or more residential units built before 1990, excluding NYCHA public housing and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Data span June 2016 through December 2025 in 6-month periods. Pooled difference-in-differences coefficients on Post × Stabilized. All outcomes are per 100 residential units, unit-weighted, except Days to Close, which is in days. Specification includes building fixed effects plus tract×period, size-bin×period, decade×period, building-type×period, and baseline-tercile×period fixed effects. "Baseline mean" is the pre-period (June 2017–May 2019) mean for stabilized buildings. Standard errors clustered at the building level in parentheses. Count-based results in Table A1.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure 3: Event Study: Effect of HSTPA on HPD Complaints



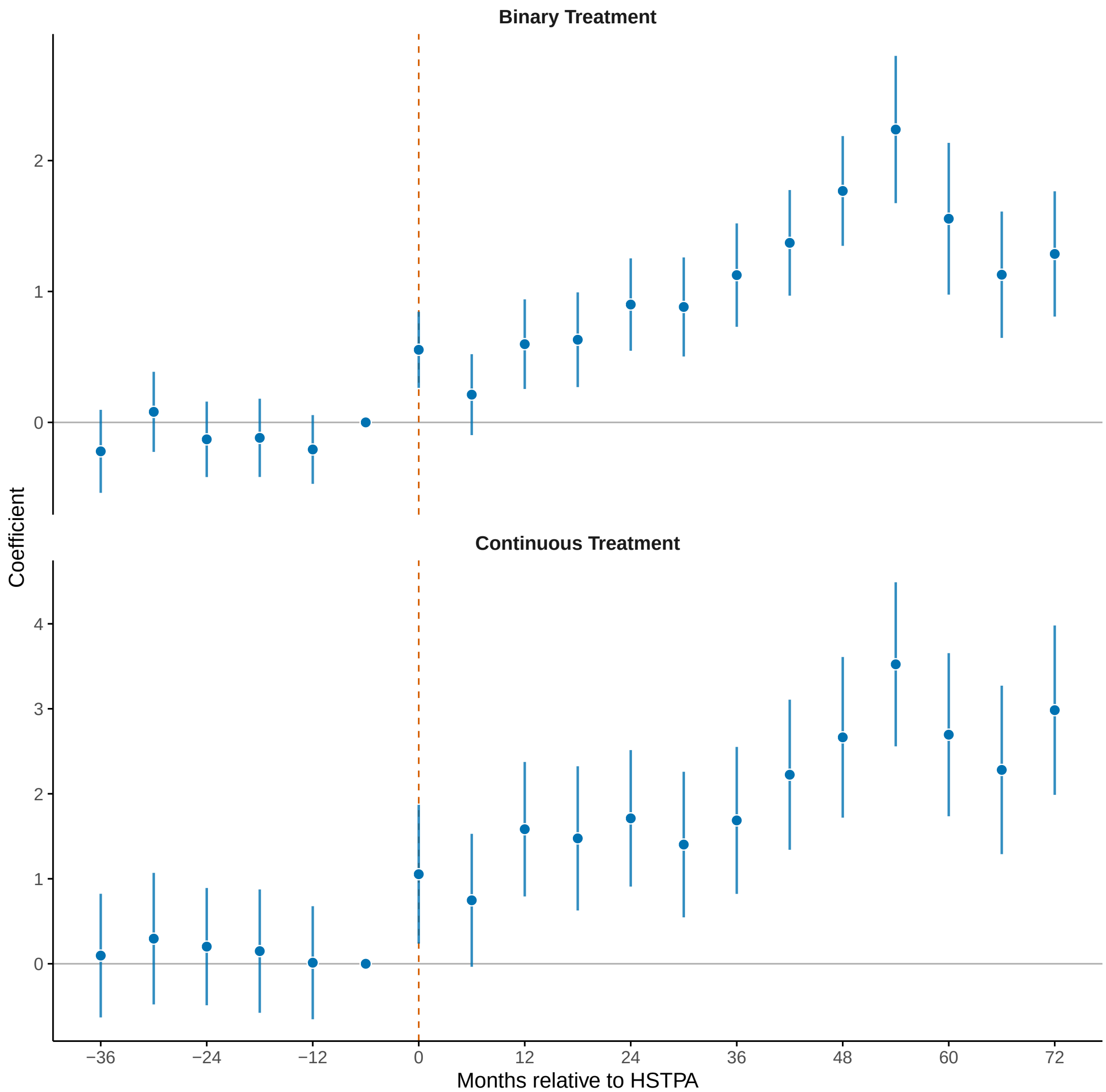
Note: Sample consists of 53,643 NYC buildings with six or more residential units built before 1990, excluding NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Figure plots coefficients on interactions between an indicator for rent-stabilized buildings and 6-month period indicators, relative to January–June 2019. Complaints are normalized to per-100-units and weighted by building size. Specification includes building fixed effects plus tract×period, size-bin×period, decade×period, building-type×period, and baseline-tercile×period fixed effects. Vertical dashed line indicates HSTPA passage (June 2019). Error bars show 95 percent confidence intervals. Standard errors are clustered at the building level.

Figure 4: Event Study: Effect of HSTPA on Building Permit Activity



Note: Sample consists of 53,643 NYC buildings with six or more residential units built before 1990, excluding NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Figure plots coefficients on interactions between an indicator for rent-stabilized buildings and 6-month period indicators, relative to January–June 2019. Outcome is DOB alteration permits (types A1, A2, A3) per 100 residential units, weighted by building size. Specification includes building fixed effects plus tract×period, size-bin×period, decade×period, building-type×period, and baseline-tercile×period fixed effects. Vertical dashed line indicates HSTPA passage (June 2019). Error bars show 95 percent confidence intervals. Standard errors are clustered at the building level.

Figure 5: Binary vs. Continuous Treatment



Note: This figure compares binary and continuous treatment specifications for Class C violations per 100 units. Sample consists of 53,643 NYC buildings with six or more residential units built before 1990, excluding condominiums. Coefficients are from 6-month period indicators interacted with treatment, relative to January–June 2019 (period $k = -1$). Top panel uses a binary indicator for rent-stabilized buildings from the 2017 DHCR registry. Bottom panel uses continuous treatment intensity (share of stabilized units, 0–1) from 2018 property tax bills. Both specifications include building fixed effects plus tract×period, size-bin×period, decade×period, building-type×period, and baseline-tercile×period fixed effects. Error bars show 95 percent confidence intervals with standard errors clustered at the building level.

Table 4: Portfolio Spillovers Test (SUTVA)

	Class C Violations
Mixed × Post	0.508*
	(0.301)
Baseline mean (mixed)	3.07
Buildings (mixed)	4,724
Buildings (all-market)	8,917
Observations	259,179

Note: This table tests whether landlords reallocate resources from stabilized to non-stabilized buildings after HSTPA (a potential SUTVA violation). Sample consists of non-stabilized buildings with identified ownership (see Appendix C). "Mixed" buildings are owned by landlords who own both stabilized and non-stabilized properties; "all-market" buildings are owned by landlords with only non-stabilized properties. The coefficient shows the differential change in Class C violations per 100 units for mixed-portfolio buildings relative to all-market buildings after HSTPA. A negative coefficient would indicate reallocation toward non-stabilized properties. Specification includes building fixed effects plus tract×period, size-bin×period, decade×period, building-type×period, and baseline-tercile×period fixed effects. Standard errors clustered at the building level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5: Heterogeneity in Class C Violation Effects

Dimension	Group	Buildings	Baseline	Effect
Overall	All	53,642	3.24	1.195***
				(0.111)
Borough	Manhattan	18,289	2.39	1.024***
				(0.095)
	Bronx	7,177	5.92	2.450***
				(0.592)
	Brooklyn	19,630	4.22	1.118***
				(0.240)
	Queens	8,146	0.95	0.976***
				(0.230)
Nbhd Income	Low	19,278	5.46	1.800***
				(0.300)
	Middle	15,397	2.88	1.378***
				(0.221)
	High	18,890	1.01	0.580***
				(0.067)
Non-white	Low	17,899	0.97	0.614***
				(0.075)
	Middle	17,922	2.52	0.842***
				(0.154)
	High	17,809	5.82	2.328***
				(0.344)
Baseline Outcome	Zero	30,910	0.00	0.990***
				(0.058)
	Low+	12,417	1.11	1.668***
				(0.206)
	High	10,315	8.78	1.554**
				(0.786)

Note: This table tests whether HSTPA's effects vary across neighborhood and building characteristics. Each row reports a separate regression on the indicated subsample. Data span June 2016 through December 2025 in 6-month periods. "Effect" is the coefficient on Stabilized×Post; "N" is the number of buildings in that subsample. Neighborhood income and non-white population share are tract-level measures from the 2019 ACS 5-year estimates (2015–2019); terciles are defined within the estimation sample. All specifications include building fixed effects plus tract×period, size-bin×period, decade×period, building-type×period, and baseline-tercile×period fixed effects. Outcome: Class C violations per 100 units, unit-weighted. Standard errors clustered at building level.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 6: Leverage Triple Difference

Leverage Group	Total Effect	(SE)	Interaction	(SE)
No mortgage	0.591***	(0.134)	—	
Low	1.102***	(0.132)	0.510***	(0.135)
Middle	2.087***	(0.171)	1.496***	(0.171)
High	1.636***	(0.240)	1.044***	(0.242)

Note: This table tests whether financial leverage amplifies HSTPA's effect on housing quality. Sample consists of NYC buildings with six or more residential units built before 1990, excluding NYCHA and subsidized housing. Data span June 2016 through December 2025 in 6-month periods. Outcome: Class C violations per 100 units, unit-weighted. "Total Effect" is the HSTPA effect (Stabilized×Post) for each leverage group. "Interaction" tests whether that group shows additional deterioration relative to buildings with no mortgage (reference). Leverage is the mortgage-to-assessed ratio using the most recent pre-HSTPA mortgage from ACRIS and 2018 assessed values from PLUTO; terciles defined among mortgaged buildings. All specifications include building fixed effects plus tract×period, size-bin×period, decade×period, building-type×period, and baseline-tercile×period fixed effects. Standard errors clustered at the building level.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

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A Additional Robustness Checks

A.1 Count-Based Results

The main text reports results normalized to per-100-units and weighted by building size. This approach accounts for heterogeneity in building size and provides a tenant-weighted interpretation: the effect represents the expected change in violations experienced by an average tenant. This appendix presents the corresponding count-based results (violations per building, unweighted), which yield qualitatively similar conclusions.

Table A1 reports the building-level count specification. The magnitude interpretation differs: while the main results show the effect per 100 residential units, the count results show the effect per building regardless of size. The patterns are consistent across specifications.

Table A1: Effect of HSTPA on HPD Violations (Count-Based)

	Class C	Class B	Complaints	Days to Close	Alterations
Stabilized × Post	0.2680***	0.1840***	0.2812***	34.52***	-0.0572***
	(0.0272)	(0.0412)	(0.0526)	(7.122)	(0.0063)
Baseline mean	0.97	2.77	3.30	100.18	0.26
Buildings	53,642	53,642	53,642	34,126	53,642
Observations	1,019,198	1,019,198	1,019,198	207,720	1,019,198

Note: This table tests whether violation results are robust to using raw counts per building rather than per-100-unit normalization. Sample consists of 53,643 NYC buildings with six or more residential units built before 1990, excluding NYCHA, subsidized housing, and condominiums. Data span June 2016 through December 2025 in 6-month periods. Each building is weighted equally (building-weighted), in contrast to the main specification where each unit is weighted equally (tenant-weighted). Specification includes building fixed effects plus tract×period, size-bin×period, decade×period, building-type×period, and baseline-tercile×period fixed effects. Standard errors clustered at the building level in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

A.2 Fixed Effects Specification

Table A2 estimates specifications with progressively richer fixed effects structures for Class C violations, demonstrating that baseline-tercile matching is essential for achieving parallel pre-trends.

Table A2: Robustness to Alternative Fixed Effects Specifications

	Bldg	+Tract	+Size	+Decade	+Class	+Baseline
Post × Stabilized	3.442***	1.477***	1.377***	1.309***	1.265***	1.195***
	(0.0665)	(0.1093)	(0.1088)	(0.1095)	(0.1110)	(0.1114)
Building	Yes	Yes	Yes	Yes	Yes	Yes
Tract × Time		Yes	Yes	Yes	Yes	Yes
Size Bin × Time			Yes	Yes	Yes	Yes
Decade Built × Time				Yes	Yes	Yes
Building Class × Time					Yes	Yes
Baseline Outcome × Time						Yes
Buildings	53,642	53,642	53,642	53,642	53,642	53,642
Pre-trend max$\|t\|$	24.97	6.180	5.600	5.580	5	1.540
Observations	1,019,198	1,019,198	1,019,198	1,019,198	1,019,198	1,019,198

Note: Sample consists of 53,643 NYC buildings with six or more residential units built before 1990, excluding NYCHA public housing, condominiums, and buildings receiving LIHTC, HUD, or Mitchell-Lama subsidies. Data span June 2016 through December 2025 in 6-month periods. Table reports pooled post-period coefficients from difference-in-differences regressions of Class C violations per 100 units (unit-weighted) on Post × Stabilized, where Post equals one for periods after June 2019. Columns differ in fixed effects structure, building progressively from building fixed effects only (column 1) to the full matching specification (column 6). "Pre-trend max$|t|$" is the maximum absolute t-statistic among pre-period event study coefficients; values above 2 indicate pre-trend concerns. Standard errors (in parentheses) are clustered at the building level.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

A.3 Continuous Treatment (Dose-Response)

This appendix presents the full dose-response results summarized in Section 5. Table A3 replaces the binary treatment indicator with the continuous share of rent-stabilized units (0–1, from 2018 property tax bills). The coefficient represents the effect of moving from 0% to 100% stabilized.

Table A4 restricts the sample to stabilized buildings only, using within-group variation in stabilization intensity as treatment.

A.4 COVID-19 Period

Table A5 compares main results with and without the COVID-19 period (January 2020 through December 2021). Results are similar across all outcomes, indicating that the main findings are not driven by pandemic-related disruptions.

Table A3: Effect of Stabilization Share on Building Outcomes

	Class C	Class B	Complaints	Permits
Stab. Share × Post	1.877***	2.181***	2.259***	-0.140***
	(0.231)	(0.319)	(0.367)	(0.039)
Baseline mean	4.70	12.26	13.01	1.14
Buildings	53,642	53,642	53,642	53,642

Note: This table tests whether effects scale with treatment intensity by using the continuous share of rent-stabilized units (0–1, from 2018 property tax bills) rather than the binary indicator. The coefficient represents the effect of moving from 0% to 100% stabilized. Sample consists of 53,643 NYC buildings with six or more residential units built before 1990, excluding NYCHA, subsidized housing, and condominiums. Data span June 2016 through December 2025 in 6-month periods. All specifications include building fixed effects plus tract×period, size×period, decade×period, building-type×period, and baseline-tercile×period fixed effects. All outcomes are per 100 residential units, weighted by building size (total units), except Days to Close. Standard errors clustered by building.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

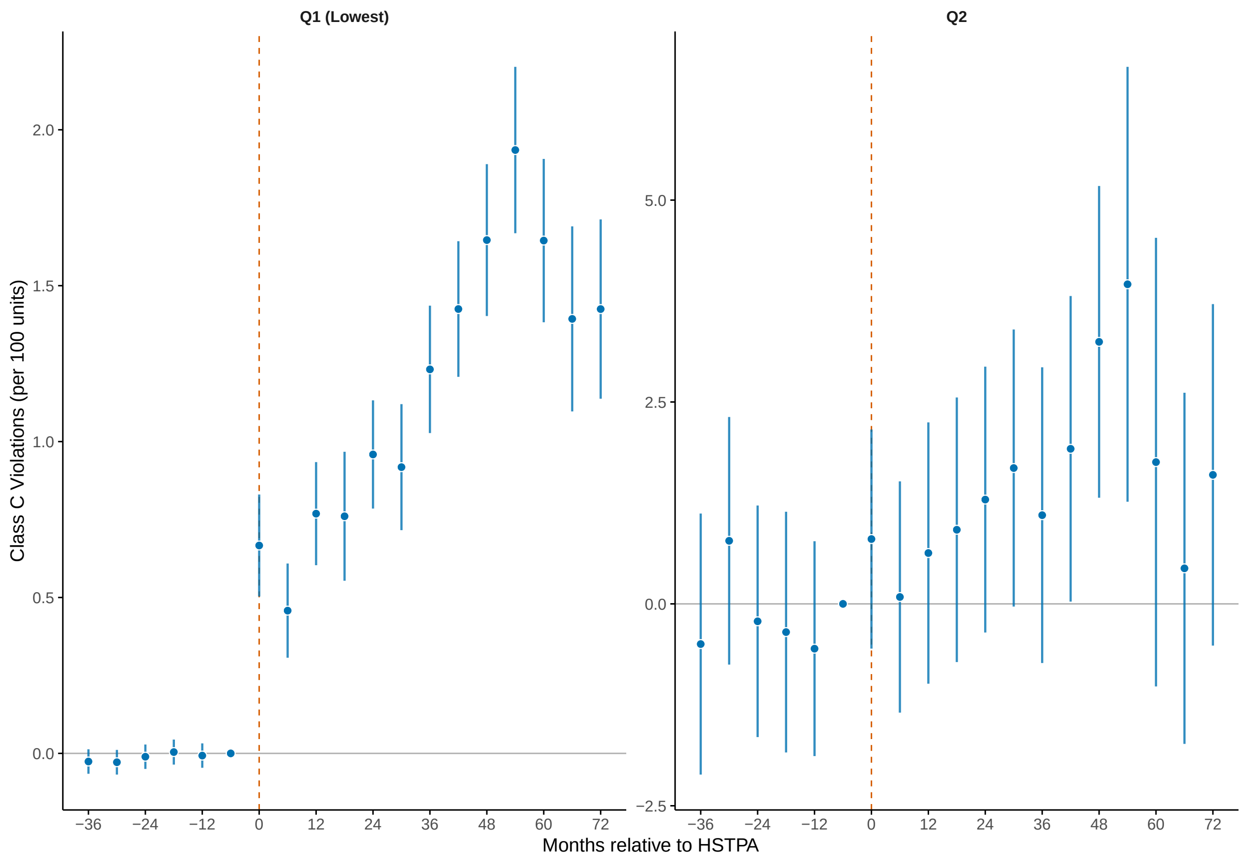
A.5 Ruling Out Mean Reversion

A potential concern with baseline-tercile matching is regression to the mean. If non-stabilized buildings in high-violation terciles are there due to transitory shocks, they may naturally revert toward lower violation rates over time. If stabilized buildings in the same terciles reflect more persistent distress, this differential mean reversion could generate a spurious positive effect even absent any impact of HSTPA.

To test for this, I estimate event studies separately within each baseline tercile. If mean reversion were driving the results, I would expect to see differential pre-trends within terciles: high-baseline non-stabilized buildings should trend downward (reverting toward the mean) relative to high-baseline stabilized buildings, generating negative pre-period coefficients in the upper terciles. Conversely, low-baseline non-stabilized buildings should trend upward, generating positive pre-period coefficients in the lower terciles.

Figure A1 presents the results. Pre-period coefficients are flat within each baseline tercile, with no evidence of differential mean reversion. Effects emerge only after HSTPA and are present across all terciles. This pattern is inconsistent with mean reversion driving the results: if differential transitory shocks were the mechanism, the divergence would appear in the pre-period, not coincident with HSTPA's passage.

Figure A1: Event Studies by Baseline Violation Tercile



Note: This figure tests whether differential mean reversion drives the results by estimating event studies separately within each baseline violation tercile. Sample consists of 53,643 NYC buildings with six or more residential units built before 1990, excluding NYCHA, subsidized housing, and condominiums. Coefficients are from 6-month period indicators, relative to January–June 2019 (period $k = -1$), for Class C violations per 100 units. T1 contains buildings with the lowest baseline violation rates; T3 contains buildings with the highest. Each specification includes building fixed effects plus tract $\times$ period, size $\times$ period, decade $\times$ period, and building type $\times$ period fixed effects (baseline-tercile $\times$ period is omitted since estimation is within-tercile). Unit-weighted. Vertical dashed line indicates HSTPA passage. Error bars show 95% confidence intervals with standard errors clustered at the building level. Pre-trends are flat within each tercile, ruling out differential mean reversion as an explanation for the main results.

Table A4: Effect of HSTPA Within Stabilized Buildings

	Class C	Class B	Complaints	Permits
Stab. Share × Post	0.921***	2.345***	1.826***	-0.064
	(0.351)	(0.489)	(0.587)	(0.051)
Baseline mean	4.70	12.26	13.01	1.14
Buildings	38,992	38,992	38,992	38,992

Note: This table restricts the sample to stabilized buildings only, using variation in the share of rent-stabilized units (from 2018 property tax bills) as a continuous treatment. The coefficient represents the effect of moving from 0% to 100% stabilized among buildings already classified as stabilized. This design eliminates the non-stabilized control group entirely, addressing concerns about control group contamination from previously decontrolled buildings. Specification otherwise identical to Table A3. All outcomes are per 100 residential units, weighted by building size (total units), except Days to Close. Standard errors clustered by building.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

B Supplementary Evidence

B.1 MCI Applications

Figure A2 shows the time series of Major Capital Improvement (MCI) applications. MCI filings collapsed immediately upon HSTPA's passage, falling from an average of 118 applications per month before HSTPA to 23 per month after, an 81 percent decline.

The June 2019 spike (313 applications, versus 118 monthly average) suggests anticipation: landlords who were planning MCI projects rushed to file before HSTPA reduced the return on such investments. The immediate collapse in July 2019 demonstrates that landlords respond quickly to changes in MCI economics.

The MCI collapse (81%) is not directly comparable to the permit decline (27%): the former is an unconditional pre/post change for stabilized buildings, while the latter is a difference-in-differences estimate relative to control buildings. The larger MCI decline likely reflects that MCIs are the rent-recovery mechanism (landlords file MCIs specifically to recoup capital costs through rent increases), while permits are just regulatory approval for construction work.

Table A5: COVID-19 Period Robustness

Outcome	Full Sample	Excl. 2020–21
Class C Violations	1.195***	1.422***
	(0.111)	(0.132)
All Complaints	1.314***	1.747***
	(0.231)	(0.261)
Days to Close	34.516***	40.799***
	(7.122)	(7.535)
Permits	-0.236***	-0.279***
	(0.029)	(0.031)
Buildings	53,642	53,642

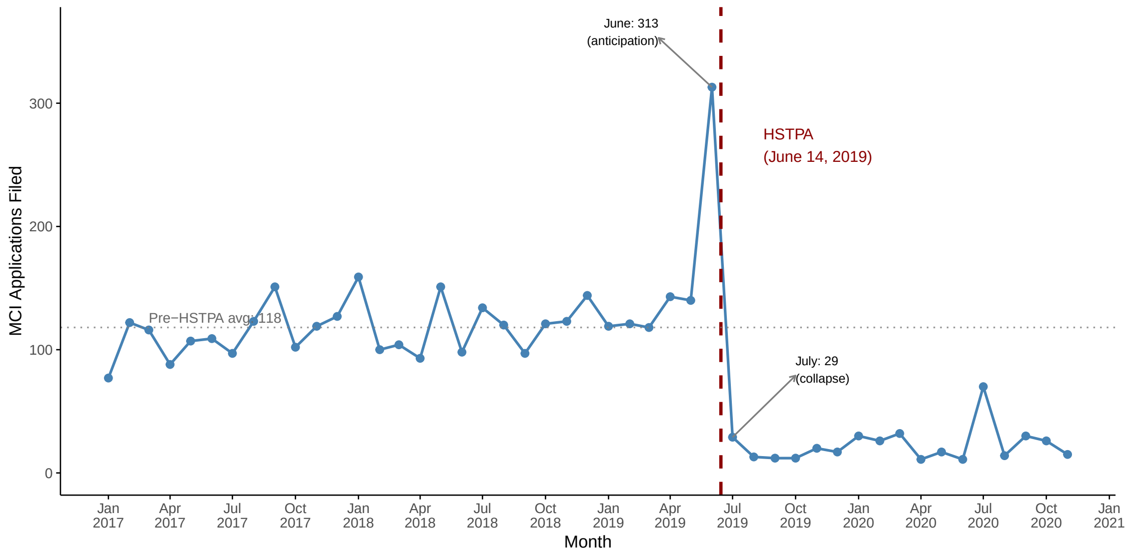
Note: Comparison of main results including vs. excluding the COVID-19 pandemic period (January 2020–December 2021). All specifications include building fixed effects plus tract×period, size-bin×period, decade×period, building-type×period, and baseline-tercile×period fixed effects. All outcomes are per 100 residential units, unit-weighted, except Days to Close. Standard errors clustered at building level.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

B.2 Violation Severity Gradient

If the violation increases reflected enhanced tenant reporting rather than actual quality decline, we would expect the largest increases for minor violations (which tenants might previously have tolerated) and smaller increases for serious violations (which tenants would report in any case). The data show the opposite pattern.

Table A6 presents effects by violation severity class. HPD classifies violations into three categories: Class A (non-hazardous, no correction deadline), Class B (hazardous, 30-day deadline), and Class C (immediately hazardous, 24-hour deadline). Class C violations increased 37% relative to baseline, Class B increased 8%, and Class A increased just 2%. This gradient is consistent with real quality deterioration concentrated in the most serious deficiencies, and inconsistent with a pure reporting-bias explanation.

Figure A2: MCI Applications Over Time



Note: Figure shows monthly Major Capital Improvement (MCI) application filings to the Division of Housing and Community Renewal (DHCR) from January 2017 through November 2020. MCIs are building-wide improvements (boilers, roofs, elevators) for which landlords of rent-stabilized buildings can seek rent increases. Vertical dashed line indicates HSTPA passage (June 14, 2019). Pre-HSTPA monthly average: 118 filings. Post-HSTPA monthly average: 23 filings (81 percent decline). The June 2019 spike (313 filings) reflects anticipation as landlords rushed to file before the law took effect. Data are HCR/DHCR application records obtained by Winnie Shen through a FOIL request and distributed through NYCDB.

C Landlord Matching and Portfolio Classification

To test for portfolio spillovers, I link buildings to individual owners using HPD Multiple Dwelling Registration data. NYC law requires owners of buildings with three or more residential units to register annually with HPD, providing contact information for the building owner. I use this data to identify which buildings share common ownership.

C.1 Data Source

The HPD Registration Contacts dataset lists contact persons for each registered building, including their role (e.g., HeadOfficer, IndividualOwner, Officer, Agent). The key challenge is that many buildings are owned through LLCs, which obscure the identity of the actual owner. However, HPD requires registration of the "head officer" or "individual owner" behind the LLC, providing person-level identifiers that can link buildings across different LLC structures.

I obtain the data from the NYC Open Data portal and link it to buildings using the HPD Multiple Dwelling Registrations dataset, which provides the BBL identifier for each registration.

Table A6: HSTPA Effects by Violation Severity Class

Class	Severity	Deadline	Effect	% of Baseline
C	Immediately hazardous	24 hours	1.20***	37%
B	Hazardous	30 days	0.70***	8%
A	Non-hazardous	None	0.08	2%

Note: This table compares HSTPA effects across violation severity classes. Sample consists of 53,643 NYC buildings with six or more residential units built before 1990, excluding NYCHA, subsidized housing, and condominiums. Effects are from pooled difference-in-differences specifications with building fixed effects plus tract×period, size-bin×period, decade×period, building-type×period, and baseline-tercile×period fixed effects. Outcomes are violations per 100 units, unit-weighted. Standard errors clustered at the building level.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

C.2 Owner Name Standardization

For each building registration, I extract the owner name using the following priority order:

1. IndividualOwner (highest priority)
2. JointOwner
3. HeadOfficer
4. Officer (lowest priority)

I construct a standardized owner identifier by concatenating first name, middle initial, and last name, converting to uppercase and removing extra whitespace. This approach links buildings owned by the same person even when held through different LLCs.

C.3 Name Deduplication

Matching on full names (first, middle, and last) avoids false positives but may fail to link the same person when names are recorded inconsistently across registrations—for example, "Steven Croman" on one building and "Steven R Croman" on another. A naive solution would merge all names sharing the same first and last name, treating middle names as optional. However, this creates substantial false positives for names where the middle component is a given name rather than an initial. For instance, "Mei Xian Chen" and "Mei Feng Chen" are distinct individuals with different given names, not spelling variants of the same person. Merging on first and last name alone would incorrectly combine thousands of distinct owners, particularly in Chinese, South Asian, and Hispanic communities where such naming patterns are common.

To balance accuracy against false positives, I implement a conservative deduplication procedure that requires direct evidence of common ownership. I identify name variants with identical first and last names that appear on the same building. When the same building lists both "Steven Croman" and "Steven R Croman" as contacts, this provides strong evidence they are the same person, and I merge all instances of both variants across the dataset. Approximately 4,400 buildings have multiple registration records, providing the overlap needed to identify such variant pairs. I also apply fuzzy string matching (requiring 85% character-level similarity) to names appearing on the same building, capturing minor spelling variations like "Hirschfield" versus "Hirscfield." This approach is conservative: it may fail to link name variants when no building happens to list both spellings.

C.4 Portfolio Classification

I merge the owner identifiers to the analysis panel and classify each owner's portfolio:

- **All-stabilized:** Owner's buildings are all rent-stabilized
- **All-market:** Owner's buildings are all non-stabilized
- **Mixed:** Owner has both stabilized and non-stabilized buildings

Table A7 summarizes the portfolio classification. Of the 24,933 unique owners identified, 1,880 (7.5%) own mixed portfolios covering 24,810 buildings. These mixed-portfolio landlords are the focus of the SUTVA test, as they are the only owners who could potentially reallocate resources between stabilized and non-stabilized properties.

Table A7: Landlord Portfolio Classification

Portfolio Type	Owners	Buildings
All-stabilized	15,228	27,227
All-market	7,825	9,021
Mixed	1,880	16,603
Total with owner info	24,933	52,851
Unknown (no owner match)	—	792

Note: This table summarizes landlord portfolio types for buildings in the analysis sample. Owner identities are derived from HPD registration contacts using the name standardization procedure described in this appendix. A portfolio is "all-stabilized" if all of an owner's buildings are rent-stabilized, "all-market" if none are stabilized, and "mixed" if the owner holds both types. Buildings that could not be linked to an owner are excluded from the SUTVA analysis (Table 4).